

Batch:	Roll No.:
Experiment / assignment / tutorial No. 01	
Grade: AA / AB / BB / BC / CC / CD / DD	
Signature of the Staff In-charge with date	

Title: Signal generation and measurement

Objectives:

1. To understand signal measurement
2. To understand effect of noise on signal

OUTCOME:

1. analyze and compare analog modulation schemes.
2. understand noise effect on the performance of communication systems.

Part-1: Signal measurement using Oscilloscope and function generator

- Visit the link: <http://vlabs.iitkgp.ac.in/aec/exp1/index.html#>
- Read the step-by-step procedure and perform the experiment
- Output: Take screenshot of one signal on the screen (of any frequency, amplitude), two signal on the screen (of any frequency and amplitude)
- Write conclusion based on your learning, measurement of parameters such as frequency, amplitude based on procedure.

Part-2: Understanding effect of noise on signal

- Open MATLAB online and sign in with your account
- Write a program to generate signal, noise and addition of signal with noise
- Copy the output waveforms of the program
- Write conclusion based on your learning and understanding of the MATLAB program

Post lab Activity:

- Watch the video on the link below and prepare concept map/mind map of entire video.
Video link: <https://www.youtube.com/watch?v=oxTUC5I22LU>
- **Reference: Watch this video (create concept map):**
<https://www.youtube.com/watch?v=sZJ6DwCqSU&t=46s>

Signature of faculty in-charge

Department of Electronics and Telecommunication Engineering (Programme: EXTC)